CS 540 Spring 2025 Final Exam

Exam Version: 1

First, Last Name:

Section:

Net ID: @wisc.edu

Campus ID Number:

1. You cannot sit next (front, back, left (within two seats including an empty seat), right (within two seats including an empty seat)) to someone you know. Switch seats now if that's the case.

2. Fill in these fields (left to right) on the scantron sheet using a pencil:

   - LAST NAME (family name) and FIRST NAME (given name), fill in bubbles
   - IDENTIFICATION NUMBER is your Campus ID number, fill in bubbles
   - Under A, B, C of SPECIAL CODES, fill in your three-digit section number (001, 002, 003)
   - **Under F of SPECIAL CODES, fill in Exam version above.** This is very important!

   Make sure you fill all the above accurately in order to get graded.

3. Mark your answers on the Scantron sheet **and this handout.** Use a #2 pencil to mark all answers.

4. You may only reference your one-sheet notes. Please turn off and put away portable electronics now.

5. If you need to use the restroom, your phone must remain in the room, placed face down on your desk and kept visible to the proctors.

6. You have 120 minutes to take the exam. Once you have finished, please show your student ID to one of the proctors and submit both your scantron and this handout.

Good luck!

1. A 2D convolutional kernel of size 3×3, stride 2×2, and padding 2×2 is applied to an input of size 9×13. What is the output size?

   A. 3×5

   B. 3×7

   C. 5×7

   D. 5×9

   E. 9×13

   Solution:

   Output dimension 1: $\frac{9-3+2+2}{2} = 5$

   Output dimension 2: $\frac{13-3+2+2}{2} = 7$

   The answer is 5x7; choice C

2. Which of the following is/are advantage(s) of convolutional neural networks?

   A. Locality

   B. Translation invariance

   C. Dense interaction

   D. Both A and B

   E. All of A, B, and C

   Solution: A and B are both correct advantages of CNNs. Choice C is incorrect; the advantage of CNNs is *sparse* interaction.

   The answer is choice D.

3. Which of the following could be used as a non-linearity in a deep neural network?

   A. Convolutional layer

   B. Max pooling

   C. Average pooling

   D. Both B and C

   E. None of the above

   Solution: Average pooling is a type of convolution, which is a linear operation. Max pooling is the only non-linear operation.

   The answer is choice B.

4. Which statement is true regarding the layer features of convolutional neural networks?
   A. Early layers recognize complex features, while later layers recognize simple features
   B. Early layers recognize simple features, while later layers recognize complex features
   C. Both early and later layers are about equally responsible for recognizing both simple and complex features
   D. There is not any observable pattern in early/later layers recognizing simple/-complex features (or vice versa)
   E. The pattern of early/later layers recognizing simple/complex features (or vice versa) varies greatly depending on the specific choice of architecture (e.g. LeNet vs. VGG vs. ResNet).

Solution: The answer is choice B

5. An RGB image of size $227 \times 227 \times 3$ is passed to a convolutional layer with 96 filters, each 11×11 in spatial extent, stride 4, no padding. Every filter has its own bias. Approximately how many learnable parameters (weights + biases) are in this layer?
   A. $\sim$ 10k parameters.
   B. $\sim$ 50k parameters.
   C. $\sim$ 35k parameters.
   D. $\sim$ 40k parameters.
   E. $\sim$ 15k parameters.

Solution: C. One filter has $11 \times 11 \times 3 = 363$ weights plus 1 bias. Hence there are 364 trainable parameters per fileter. Hence, the total number of trainable parameters are $364 \times 96 = 34944 \sim 35$k.

6. Which architectural change can help with the vanishing-gradient problem?
   A. Adding more layers
   B. Training on two GPUs in parallel
   C. Using aggressive data augmentation
   D. Replacing sigmoids with ReLU activations
   E. Setting a larger learning rate

Solution: D. The non-saturating nature of ReLU ($\max(0, x)$) helps prevent gradients from shrinking, unlike sigmoids.

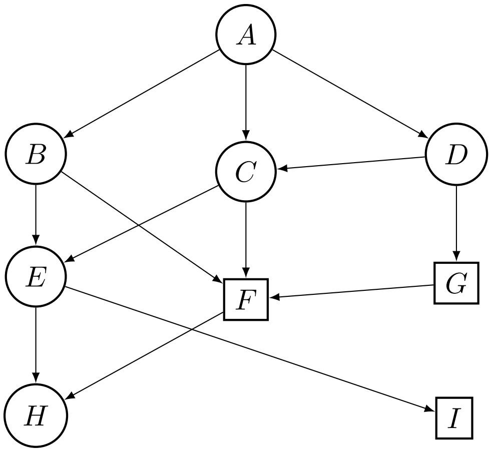
7. Which of the following is true about residual networks?
   A. Residual connections can avoid the vanishing gradient problem by allowing parts of the network to learn the identity function.
   B. Residual connections result in improved performance at the cost of requiring additional parameters to learn.
   C. Residual connections can avoid the vanishing gradient problem by allowing parts of the network to learn the zero function.
   D. Residual networks are the only types of neural networks where batch normalization can be used.
   E. None of the above.

Solution: C: the identity function is hard to learn, so residual connections allow parts of the network to learn the zero function due to the skip connections. The zero function is easier to learn since all weights can be set to be very small or zero. A is false since residual connections are used to avoid needing to learn the identity function in large networks. B is false since residual connections do not require additional parameters. D is false since batch normalization is a common trick used in deep learning in general.

8. You have started creating a deep learning model on image data. However, you notice your model has low training error but high generalization error. Which of the following is **not** a valid way of improving your model performance?
   A. Modify the loss function with a regularizer and regularization parameter.
   B. Use data augmentation to expand your dataset by slightly changing its color.
   C. Add additional layers to your neural network so it can learn a more expressive function.
   D. Apply batch normalization.
   E. None of the above

Solution: C. Your model already has low training error (indicating that it has learned well) but does not generalize well, which implies it is overfitting. Adding additional layers, or increasing model complexity, is unlikely to improve performance. A: adding regularization is a good idea to avoid overfitting. B: data augmentation is a good idea but adding some noise can decrease dependence on the existing images. D: Batch normalization can help improve overfitting.

9. In the graph below, **goal states are in rectangles**. Which goal state will be found and returned by DFS and BFS? Break ties by alphabetical order (ie. letters that come earlier in the alphabet get higher priority).



A. BFS: F, DFS: F

B. BFS: F, DFS: I

C. BFS: I, DFS: F

D. BFS: F, DFS: G

E. BFS: G, DFS: I

Solution: B: In BFS, we process the nodes in order A,B,C,D,E,F which is a goal state. In DFS, we process the nodes in order A,B,E,H,I which is a goal state.

10. In Uniform Cost Search (UCS), which of the following data structures is typically used to keep track of the nodes to be visited next?

A. Stack

B. Queue

C. Priority queue

D. All the above

E. None of the above

Solution: C

11. Let h_0 be an admissible heuristic for a search problem. Which of the following h will **always** be admissible?

A. $h(s) = \sqrt{h_0(s)}$

B. $h(s) = (h_0(s))^2$

C. $h(s) = h_0(s) - 1$

D. $h(s) = h_0(s) + 1$

E. None of the above.

Solution: The correct choice is **E, none of the above**. Consider the case where $h_0(s) = h^*(s)$. Choice A is incorrect because if $h^*(s) < 1$ then $\sqrt{h^*(s)} > h^*(s)$. Choice B is incorrect because if $h^*(s) > 1$ then $(h^*(s))^2 > h^*(s)$. Choice C is incorrect because if $h^*(s) < 1$ then $h^*(s) - 1 < 0$. Choice D is incorrect because $h^*(s) + 1 > h^*(s)$.

12. Consider the following search algorithms:

(i) Uniform Cost Search

(ii) Depth First Search

(iii) A search

(iv) A^* search

Which of the above algorithms are both complete and optimal?

A. All of them.

B. (iv) only.

C. (i) and (iv) only.

D. (iii) and (iv) only.

E. (i), (iii), and (iv) only.

Solution: The correct choice is **C, (i) and (iv) only**. Depth First Search is neither complete nor optimal, and A search is not optimal when its heuristic is not admissible.

13. Consider a "flower" graph with $3n+1$ nodes $V = \{0, 1, \ldots, 3n\}$ and edges $E = \{(0, i) | 1 \leq i \leq n \text{ and } 2n + 1 \leq i \leq 3n\} \cup \{(n - 4, n + 1)\} \cup \{(i, i + 1) | n + 1 \leq i \leq 2n - 1\}$ where $n \geq 4$ is an integer. The starting node is 0 and the goal is node $2n$. When two nodes are tied for priority to be explored next, choose the one of smaller numerical value. Suppose $n = 43$. Let a be the number of nodes left in the fringe when the goal is examined for BFS and b be the number of nodes left in the fringe when the goal is examined for DFS. What is $b - a$?

A. 0

B. 39

C. 43

D. 47

E. 86

Solution: BFS will first examine node 0, then all of the petals followed by the stem, with the goal node remaining the sole un-examined node immediately before exploring it. This leaves a fringe with size 0. DFS will examine nodes $0, 1, \ldots, n-4$, then proceed down the stem until examining the goal, leaving $n-3, n-2, n-1, n, 2n+1, \ldots, 3n$ in the fringe at termination. There are $4 + n = 47$ such nodes.

14. Consider the following simultaneous-move game between players 1 and 2 with payoff matrix:

$Player1 \backslash Player2$	L	R
U	$(4,1)$	$(2,2)$
D	$(3,3)$	$(1,4)$

Which strategy is strictly dominant for player 2?

A. L

B. R

C. Both L and R

D. Neither L nor R

E. It depends on player 1's action

Solution: B. Compare player 2's payoffs: - If player 1 plays U: $u_2(L) = 1$, $u_2(R) = 2$. - If player 1 plays D: $u_2(L) = 3$, $u_2(R) = 4$. In both cases $u_2(R) > u_2(L)$, so R strictly dominates L. Hence the strictly dominant strategy for player 2 is R.

15. Consider a sequential game where Player A moves first and has 3 possible actions. At each of A's nodes, Player B then has 2 possible actions; at each of those nodes, Player A again has 3 actions; and finally Player B has 1 action at each resulting node. How many nodes are in the full minimax tree (including all internal and leaf nodes)?

A. 28

B. 46

C. 36

D. 42

E. 50

Solution: The total nodes is

$$1 \; + \; 3 \; + \; 3 \times 2 \; + \; 3 \times 2 \times 3 \; + \; 3 \times 2 \times 3 \times 1 \; = \; 1 + 3 + 6 + 18 + 18 \; = \; 46.$$

Hence the answer is 46.

16. Which of the following statements about a Markov Decision Process (MDP) is not compatible with the Markov assumption?

    A. The transition probability depends only on the current state and action.

    B. The reward function outputs a scalar value.

    C. The transition probability depends on current and previous states and actions.

    D. The policy maps each state to an action.

    E. The goal is to find a policy that maximizes cumulative reward.

Solution: The Markov assumption requires that $P(s_{t+1} \mid s_t, a_t)$ depends only on (s_t, a_t), not on earlier history, so option C is false.

17. Assume you are running a Q-learning algorithm with ϵ-greedy policy. The update rule for the Q-value of a state-action pair (s, a) is given by:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s,a)]$$

Where α is the learning rate, γ is the discount factor, r is the reward, s' is the next state, and a' represents possible actions from s'.

Which of the following statements is true?

    A. During exploration we always take the action with the highest $Q(s,a)$ value.

    B. Higher learning rate means that the effect of the old $Q(s,a)$ value is more significant, while lower learning rate means that the effect of the old $Q(s,a)$ value is less significant.

    C. It requires a model of the environment's transition probabilities to update Q-values.

    D. By taking a random action with probability ϵ we might get a more accurate model of the environment.

    E. None of the above

Solution: The correct answer is **D**.

18. Supposed you have the following environment where a robot can move:

s_0	s_1	s_2
s_3	s_4	s_5
s_6	s_7	s_8

The robot can only move horizontally or vertically by taking the following actions u (up), d (down), l (left), r (right) and you have the following information:

1. The reward in s_3 is 2
2. The transition probabilities are: $P(s_0|s_3, r) = 0.1$ and $P(s_4|s_3, r) = 0.7$ and $P(s_6|s_3, r) = 0.2$
3. Currently, $V(s_0) = 0$ and $V(s_4) = 10$ and $V(s_6) = 5$
4. The discount factor is 0.5

Remember the update for value iteration is $V_{t+1}(s) = r(s) + \gamma \max_a \sum_{s'} P(s'|s, a)V(s')$. Assuming the maximizing action taken from s_3 is r, what is the value $V(s_3)$ for state s_3, after a single iteration of value iteration?

A. 1

B. 2

C. 4

D. 5

E. 6

Solution: E

$V(s_3) = 2 + 0.5 \times (0.1 \times 0 + 0.7 \times 10 + 0.2 \times 5) = 6$

19. Which of the following is true about neural networks **generalizing** to test data?

A. For a classification task, a neural network with high prediction accuracy on training data **will have** high prediction accuracy on the test data.

B. For a classification task, a neural network with high prediction accuracy on training data is **likely to have** high prediction accuracy on the test data.

C. Regularization techniques will make generalization worse.

D. Any neural network that does not have a vanishing gradient problem can generalize well.

E. Training a model with less data can improve generalization.

Solution: B

20. In simulated annealing, the algorithm accepts worse moves with a probability that depends on a temperature parameter. What is the most likely outcome if the temperature is cooled too quickly or set too low early in the process?

A. The algorithm will behave like a random search, accepting or rejecting moves with equal likelihood.

B. The algorithm will behave similarly to hill climbing, virtually never accepting worse moves and becoming prone to getting stuck in local optima.

C. The algorithm will frequently accept worse moves, causing it to oscillate erratically and delay convergence.

D. The algorithm will maintain a robust exploration of the search space by dynamically increasing the chance of accepting worse moves.

E. The algorithm will immediately converge to the global optimum because the low temperature forces it to exploit the best nearby solution.

Solution: Correct Answer: B

Explanation: Simulated annealing accepts worse moves with a probability proportional to $e^{\frac{-\Delta E}{T}}$, where ΔE is the increase in the cost function and T is the temperature. When the temperature T is too low, the acceptance probability for worse moves becomes nearly zero. As a result, the algorithm only accepts moves that improve or leave unchanged the current solution, effectively behaving like hill climbing. This makes it more susceptible to being trapped in local optima and less capable of exploring the solution space globally.

21. Imagine you are developing a navigation system for a delivery robot operating in a static warehouse. The layout of the warehouse is fixed and well-known, and an efficient heuristic (such as Manhattan distance) can be used to estimate the shortest path between two points. You are considering different search algorithms to compute the optimal route. Which search algorithm can provide the optimal path for the robot to its target?

A. Hill climbing

B. Simulated annealing

C. Depth First Search (DFS)

D. Breadth First Search (BFS)

E. A^* search

Solution: Correct Answer: E

22. Hill climbing and stochastic gradient descent are related in which of the following ways?

A. Both require computing a gradient to find better states.

B. Both seek better solutions by iteratively moving to improving states.

C. Both always find global optima in any landscape.

D. Both avoid local optima through backtracking.

Solution: B.

Gradient requirement: only SGD requires this.

Global optimum: not guaranteed unless the function is convex.

Backtracking: not a core feature of either algorithm.

23. You are tasked with reducing the pre-existing biases of an ML model and to increase human trust in the model's predictions. What is/are some technique(s) you can try?
    A. Address *class imbalance* by collecting more data for under-represented groups in the training data.
    B. Remove sensitive attributes from the training data (race, gender, income).
    C. Address *spurious correlations* by adding "counterfactuals" to the training data, such as reversing gender roles.
    D. Remove data that can be used as proxies for sensitive information (e.g., school district).
    E. All of the above

Solution: E is correct, as all of the options are techniques to address bias and increase trust (A, C, D are from lecture slides).

24. Which of the following is true about training neural networks?
    A. Deep neural networks are trained with nonlinearities like Sigmoid and ReLU as activation functions.
    B. High training error indicates that the neural network is overfitting.
    C. A purely linear neural network can get 100% accuracy on the XOR classification task.
    D. When computing resources are limited, gradient descent is preferred over mini-batch stochastic gradient descent.
    E. None of the Above.

Solution: **A** is correct.

25. Suppose you flip a biased coin 10 times and observe 6 Heads and 4 Tails. What is the maximum likelihood estimate (MLE) for the probability θ of Heads?
    A. $\theta = 0.4$
    B. $\theta = 0.5$

C. $\theta = 0.6$

D. $\theta = 0.67$

> **Solution:** C.
> $$\theta^* = \frac{\#\text{ Heads}}{\text{Total trials}} = \frac{6}{10} = 0.6$$

26. Given a set of data points $(x_0, y_0) = (\{1, 0\}, 2)$, $(x_1, y_1) = (\{1, -1\}, 1)$ and $(x_2, y_2) = (\{1,\ 1\}, \text{-}1)$. Let $w = (w_1, w_2) \in \mathbb{R}^2$ be the weights (assume bias $b = 0$) computed on performing least squares using linear regression on this dataset. Then the value of $w_1 w_2$ is?

    A. $-\frac{2}{3}$

    B. 4

    C. 1

    D. -1

    E. $\frac{5}{3}$

> **Solution:** Given the data points $X = \{(x_0, y_0), (x_1, y_1), (x_2, y_2)\}$, we can write the LSQ objective as:
>
> $$\begin{aligned} L &= (2 - w_1)^2 + (1 - w_1 + w_2)^2 + (-1 - w_1 - w_2)^2 && (1) \\ &= 3w_1^2 + 2w_2^2 - 4w_1 + 4w_2 + 6 && (2) \end{aligned}$$
>
> Now setting the derivates $\frac{\partial L}{\partial w_1} = 0$ and $\frac{\partial L}{\partial w_2} = 0$ we get,
>
> $$w_1 = \frac{2}{3} \text{ and } w_2 = -1 \qquad (3)$$
>
> The value of $w_1 w_2 = -\frac{2}{3}$. The correct answer is (A)

27. You have five 2-D points $(0, 0), (2, 0), (0, 2), (2, 2), (5, 5)$. You run 2-means with initial centroids $C_1 = (0, 0)$, $C_2 = (5, 5)$. After one assignment-and-update step, what are the new centroids?

    A. $C_1 = (0.5, 0.5)$, $C_2 = (4, 4)$

    B. $C_1 = (2, 2)$, $C_2 = (5, 5)$

    C. $C_1 = (1, 1)$, $C_2 = (5, 5)$

    D. $C_1 = (1.5, 1.5)$, $C_2 = (5, 5)$

    E. $C_1 = (1, 1)$, $C_2 = (4.5, 4.5)$

Solution: Points $(0,0),(2,0),(0,2),(2,2)$ all go to C_1, whose mean is $(1,1)$. Point $(5,5)$ stays at C_2.

28. The cross entropy loss function is defined as:

$$L_{CE} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

Where y_i represents the true probability distribution (usually one-hot encoded labels), $\hat{y}_i$ represents the predicted probability distribution, and C is the number of classes.

Which of the following **BEST** describes when to use cross entropy loss and its primary advantage?

A. It is primarily used for regression problems and its main advantage is that it penalizes predictions proportionally to their distance from the true value.

B. It is primarily used for classification problems and its main advantage is that it heavily penalizes confident but incorrect predictions.

C. It is primarily used for clustering problems and its main advantage is that it minimizes the distance between cluster centers.

D. It is primarily used for classification problems and its main advantage is that it converges faster than other loss functions regardless of prediction confidence.

E. All the above.

Solution: The correct answer is **B**. The other choices describe the properties of other loss functions.

29. Which of the following statements about gradient descent is **CORRECT**?

A. Gradient descent is guaranteed to find the global minimum of any objective function if the learning rate is sufficiently small.

B. The learning rate α should ideally be set very large to ensure rapid convergence to the optimal solution.

C. For convex problems (where all optima have the same value), gradient descent will converge to the global minimum given an appropriate learning rate.

D. Minibatch Stochastic Gradient Descent iterates over all the data points of the dataset in one iteration.

E. During gradient descent the model parameters are fixed and the model output is updated according to the input.

Solution: The correct answer is **C**.

30. Which of the following statements is true about Nash equilibrium?
    A. In Mixed Strategy Nash Equilibrium all players choose their actions deterministically.
    B. Every finite (players, actions) game has at least one pure Nash equilibrium.
    C. A pure Nash Equilibrium is also always a Dominant Strategy Equilibrium.
    D. A Dominant Strategy Equilibrium is also always a pure Nash Equilibrium.
    E. All of the above.

Solution: The correct answer is **D**. A DSE is also a pure Nash Equilibrium, all the rest are not correct.